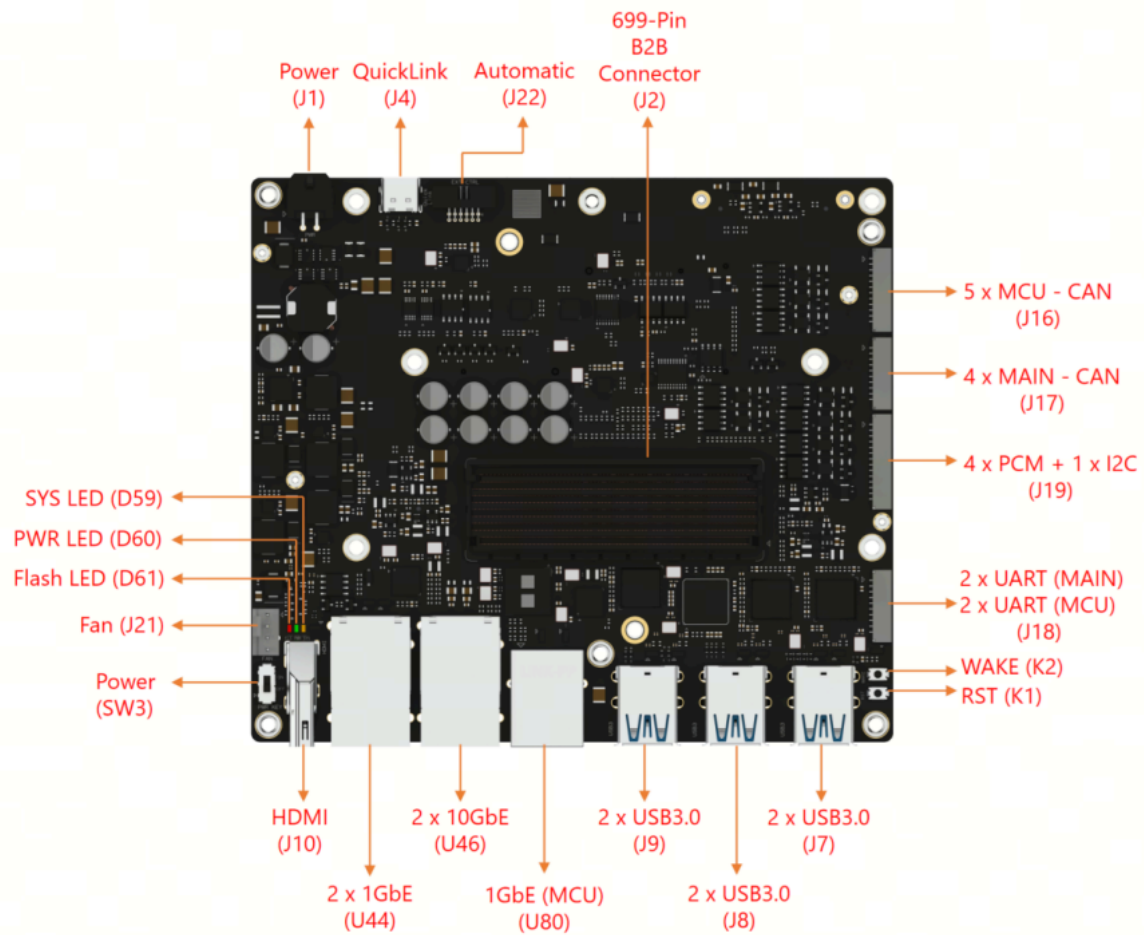


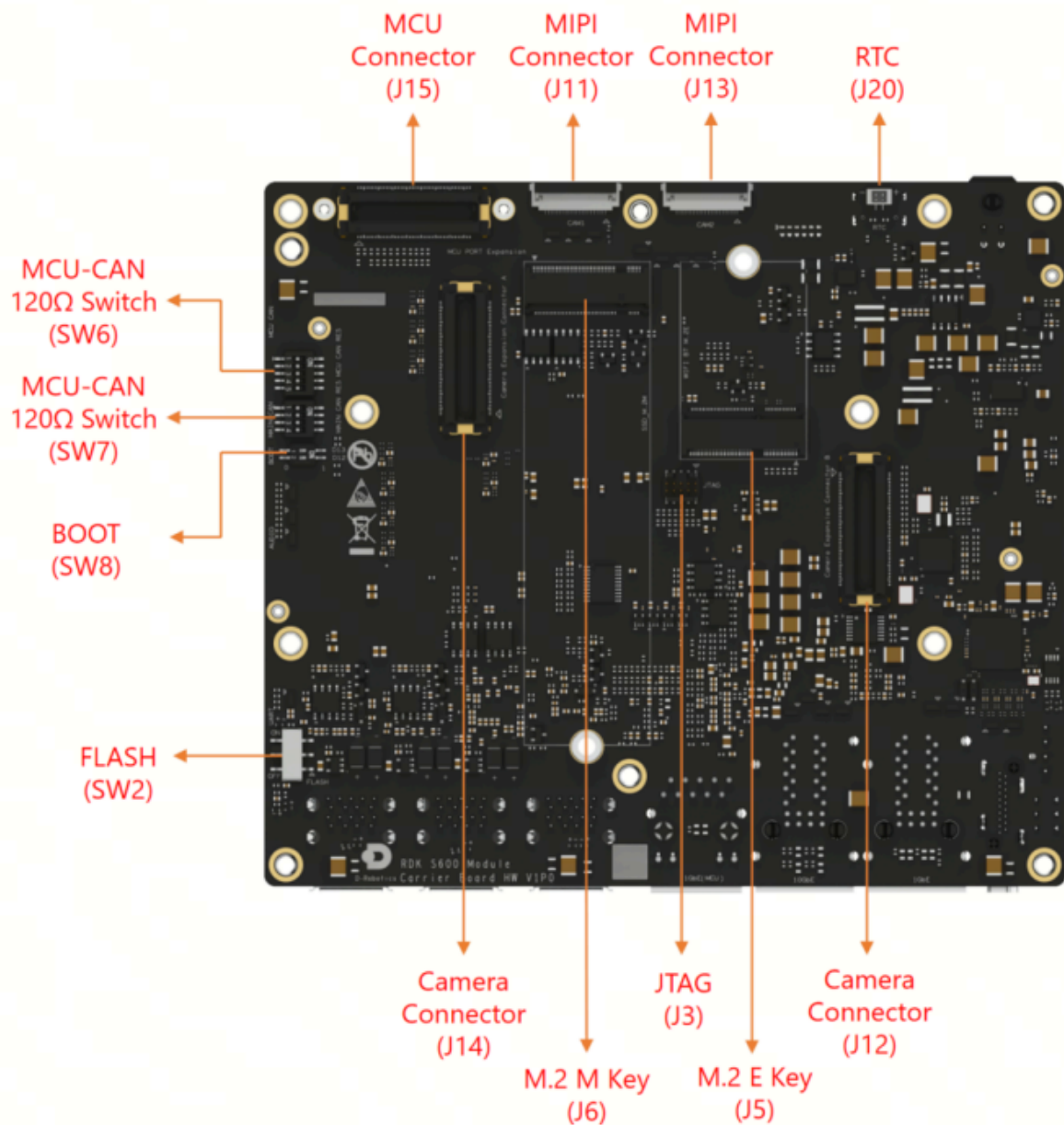
Hardware Interface Description

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1. Interface Layout





No.	Function	No.	Function	No.	Function
D59	System LED	J11	MIPI Camera Interface	J22	Main Board Function Interface
D60	Power LED	J12	Camera Expansion Interface	K1	RST Button
D61	Flash LED	J13	MIPI Camera Interface	K2	WAKE Button
J1	Main Board Power Interface	J14	Camera Expansion Interface	SW2	Flash Switch
J2	699-Pin B2B Connector	J15	MCU Expansion Interface	SW3	Power Switch
J3	JTAG Interface (MCU&MAIN)	J16	MCU-CAN Interface	SW6	MCU-CAN Resistor Selection
J4	Flash (Flashing, Main&MCU Debug)	J17	MAIN-CAN Interface	SW7	MAIN-CAN Resistor Selection
J5	M.2 Key E Interface	J18	UART Interface (MAIN&MCU)	SW8	BOOT
J6	M.2 Key M Interface	J19	PCM+I2C Interface	U44	2x 1GbE
J7/J8/J9	USB 3.0 Type-A Interface	J20	RTC Battery Interface	U45	2x 10GbE
J10	HDMI Interface	J21	Fan Interface	U80	1x 1GbE (MCU)

2. Main Board Power Interface

- The RDK S600 Developer Kit supports 12V-28V power supply, with a maximum current of 16A, using a 4-pin Microfit Connector.
- D-Robotics provides a power adapter with a rated output voltage of 24V and a maximum output current of 8A.

3. Type-C Interface

The USB Type-C (J16) is used only for flashing and debugging. It is not a standard full-featured USB Type-C port and includes the following functions:

- Debug Serial Port. Hardware-wise, two `CH340` chips convert the debug serial ports of the Main domain and MCU domain of the compute module into USB interfaces. Users can use this interface for various debugging tasks. The first time you use this interface, you need to install the CH340 driver on your computer. You can search for the keyword `CH340 serial driver` to download and install it. The parameters of the computer's serial tool need to be configured as follows:
 - Baud rate: 921600
 - Data bits: 8
 - Parity: None
 - Stop bits: 1
 - Flow Control: None
- USB Download Interface. The download interface provided by the RDK S600 development board is used for firmware download. For details, please refer to [1.2 System Flashing]

4. HDMI Interface

The RDK S600 Developer Kit provides one HDMI display interface, supporting display modes up to 2k at 60 frames per second. Upon power-on, the developer kit outputs the Ubuntu graphical interface via the HDMI interface. With specific example programs, it also supports preview display of camera and video stream feeds.

5. USB 3.0 Type-A Interface

The RDK S600 development board provides six USB 3.0 standard interfaces extended via PCIe, capable of supporting six USB peripherals simultaneously. Each USB 3.0 interface supports a maximum output power of 5V/1A.

6. Fan Interface

The FAN Connector is used to connect a cooling fan and supports fan speed control.

Note: The maximum supply current for 12V is 200mA. When the system is in light sleep and deep sleep modes, the VDD_PERI_12V power supply is turned off.

7. M.2 Key E Interface

Default configuration for connecting a PCIe Wi-Fi and USB 2.0 BT module, also supports UART 4-wire BT modules. PCIe supports Gen3x1. Also provides SDIO interface, PCM audio interface, no LED or ALERT signals.

- If support for Wi-Fi/Bluetooth sleep/wake-up of the S600 is required, please contact D-Robotics engineers for support.
- When the system is in light sleep and deep sleep modes, the VDD_AON_PERI_3V3 power supply remains on, with a maximum supply current of 750mA.

8. M.2 Key M Interface

Used for connecting a PCIe solid-state drive (SSD), supports Gen4x2, has an ALERT signal, no LED signal.

Note:

- Sleep/wake-up functionality is not supported.
- When the system is in light sleep and deep sleep modes, the VDD_PERI_3V3 power supply is turned off, with a maximum supply current of 700mA.

9.MCU Expansion Interface

The RDK S600 Developer Kit includes an 80-Pin expansion interface, referred to as the MCU 80-Pin Connector.

- Power signals: VDD_MCU_PERI_5V, VDD_MCU_PERI_3V3, VDDIO_MCU_3V3, VDDIO_MCU_1V8
- Communication interfaces:
 - I2C13, configured with an equivalent 2.2K pull-up resistor on the mainboard.
 - I2C14, configured with an equivalent 2.2K pull-up resistor on the mainboard.
 - SPI4, SPI6, SPI8, SPI13.
- CAN interfaces: CAN1, CAN2, CAN3, CAN4, CAN10
- Other signals: GPIO, PWM, ADC, PPS signal

Note:

- The MCU 80-Pin Connector includes IOs with both 1.8V and 3.3V logic levels. For detailed information, please refer to interface description.
- When the system is in light sleep, the VDD_MCU_PERI_5V and VDD_MCU_PERI_3V3 power supplies remain on, with a maximum supply current of 1000mA each.
- When the system is in deep sleep, the VDD_MCU_PERI_5V and VDD_MCU_PERI_3V3 power supplies are turned off.
- VDDIO_MCU_3V3 and VDDIO_MCU_1V8 have a maximum supply current of 50mA each, suitable only for low-power applications such as level shifting, pull-up/down resistors, ADC voltage dividers, etc. When the system is in light sleep mode, VDDIO_MCU_3V3 and VDDIO_MCU_1V8 remain on. When the system is in deep sleep mode, VDDIO_MCU_3V3 and VDDIO_MCU_1V8 are turned off.

10. Camera Expansion Interface

The RDK S600 Developer Kit includes 2 camera expansion interfaces. The camera expansion interface uses an 80-pin board-to-board connector. Each camera expansion interface can provide the following functions. See [Interface Description] for camera expansion interface signal definitions.

- 2x D-PHY or configurable to 1x D-PHY + 1x C-PHY.
- 2x IIC interfaces.
- 4x PWM signals for camera triggering.
- 2x sensor error signals.
- 2x deserializer lock signals.
- 2x deserializer error signals.
- 2x deserializer PWRON signals.
- 2x POC EN signals.
- 2x POC INT signals.
- 12V power supply, max 2A.
- 3.3V power supply, max 1A.
- 1.8V power supply, max 50mA, used for signal pull

Note:

- The maximum power supply capability for 12V is 2A, please pay attention to the total power consumption of the sensors used.

11.Ethernet Port

2x 1GbE (U44)

Two Gigabit Ethernet interfaces, compatible with 1000BASE-T and 100BASE-T standards, supporting auto-negotiation speed switching.

Position	Identifier	Function Description	IP Configuration Method	Default IP Address
Lower	eth0	General-purpose Ethernet, user needs to configure IP	External DHCP assignment or manual static config	None
Upper	eth1	Management or dedicated communication interface, built-in static IP	Fixed static IP	192.168.127.10

2x 10GbE (U45)

This interface provides 2 10-Gigabit Ethernet ports.

1x 1GbE MCU (U80)

This interface is the MCU domain Gigabit Ethernet port.

12.Switch, Button, and LED Description

Indicator LEDs (D59/D60/D61)

Reference	Name	Off (State)	On (State)	Blinking (State)
D59	System LED	System abnormal	System abnormal	System running normally
D60	Power LED	Power off	Power on	-
D61	Flash LED	Normal boot mode	DFU flashing mode	-

Switches (SW2/SW3)

Reference	Name	OFF	ON	Notes
SW2	Flash Switch	Normal boot mode	DFU flashing mode	-
SW3	Power Switch	Power off	Power on	[△] position is ON

DIP Switches (SW6/SW7)

位号	名称	OFF	ON	Notes
SW6	MCU-CAN1 120Ω	OFF	ON	-
	MCU-CAN2 120Ω	OFF	ON	-
	MCU-CAN3 120Ω	OFF	ON	-
	MCU-CAN4 120Ω	OFF	ON	-
	MCU-CAN5 120Ω	OFF	ON	-
SW7	MAIN-CAN1 120Ω	OFF	ON	-
	MAIN-CAN2 120Ω	OFF	ON	-
	MAIN-CAN3 120Ω	OFF	ON	-
	MAIN-CAN4 120Ω	OFF	ON	-

BOOT (SW8)

For SW8 DIP switch, set to 1 for ON, set to 0 for OFF.

Reference	D12	D13	Function
SW8	OFF	ON	System image boots from NVMe
	ON	OFF	-
	ON	ON	System image boots from UFS
	OFF	OFF	System image boots from UFS

Buttons (K1/K2)

Reference	Name	Description	Operation Method
K1	RST Button	Resets the S600 system	Pressing the button resets the system
K2	WAKE Button	Controls S600 system sleep or wake-up	Single-press button interrupt triggers sleep and wake-up functions